Formulario Termo

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Adiabatic procces $q = 0$

Work of reversible adiabatic expansion

$$w_{\text{ad}} = n\overline{\mathrm{C}}_v \Delta T = \Delta u$$

In a monoatomic gas $C_v = \frac{3}{2}R$ and $C_p = \frac{5}{2}R$, in a diatomic gas $C_v = \frac{5}{2}R$ and $C_p = \frac{7}{2}R$,
Temperature change, reversible adiabatic expansion

$$T_f = T_i \left(\frac{V_i}{V_f}\right)^{\frac{1}{c}}$$

where $c = \frac{C_{v,m}}{R}$
Pressure change, reversible adiabatic expansion

$$V_i T_i^c = V_f T_f^c$$

Pressure change

$$P_f V_f^\gamma = P_i V_i^\gamma$$

where $\gamma = \frac{C_{p,n}}{C_{v,n}}$

$$T_i V_i^{\gamma-1} = T_f V_f^{\gamma-1}$$

Segunda Ley

La entropía viene de aquí, nos permite conocer la dirección de la reacción

$$\Delta S = \frac{q_{\text{rev}}}{T}$$

A gas expands isothermally

The reversible work in an **isothermal expansion** is $w_{\text{rev}} = -nRT\ln\left(\frac{V_f}{V_i}\right)$, hence because for this isothermal process of a perfect gas q = −w, we have:

$$\Delta S = nR\ln\left(\frac{V_f}{V_i}\right)$$

here the change is isothermal so we could use $P_i V_i = P_f V_f$
We could alos use hess law to calculate the entropy ΔS
Entalpy in physical procces $\Delta H = nC_{p,n}$

If expansion is irreversible (spontaneous) then $\Delta S_{\text{universe}} > 0$ where $\Delta S_{\text{univerese}} = \Delta S_{\text{system}} + \Delta S_{\text{wall}}$

Avogadros principle

here the entropyof a gas mix steps to calculate it, with formulas and explicilty explainin each one, with $X_a = \frac{n_a}{n_a = n_b}$ and how this translates to the diff in entropy $\Delta S = -R(n_a \ln(X_a) + n_b \ln(X_b))$ the entropy of a mix will be positive

change in entropy by heating

$$\Delta S = nC_{p,n} \ln\left(\frac{T_2}{T_1}\right)$$

where $T_2 > T_1$

Carnot Cycle

We have 4 procceses four reversible stages in which a gas (the working substance) is either expanded or compressed in a par- ticular sequence of ways (Fig. 3A.8). In stage 1 heat is trans- ferred from a hot source to the gas, and in stage 3 energy as heat is transferred from the gas to a cold sink. Stage 1 is the isothermal reversible expansion at the temperature Th. Stage 2 is a reversible adiabatic expansion in which the temperature falls from Th to Tc. Stage 3 is an isothermal reversible compression at Tc. Stage 4 is an adiabatic reversible compression, which restores the system to its initial state.
ADD

$$\Delta S = q_1 + q_2 + q_3 + q_4 = q_1 + q_3$$

where $q_1 > 0$ and $q_3 < 0$ for a perfect gas the entropy change is zero

STEPS TO CSLCULATE A WHOLE CARNOT Cycle

Eficiency e

The eficiency is always positive and is

$$e = 1 + \frac{q_1}{q_3}$$

also

$$e = 1 - \frac{T_{\text{low}}}{T_{\text{high}}}$$

$$e = -\frac{w}{q}$$

Solving for carnot

- Step 1 isothermal

$q_1 = -w_1 = nRT_{\text{high}} \ln\left(\frac{V_2}{V_1}\right)$

- Step 2 adiabatic

$w_2 = nC_{v,n}\Delta T$ where $\Delta T = T_L - T_H$
$q_2 = 0$

- Step 3 isothermal

complete

- Step 4 adiabatic (formulas above of Temp, pressure and volume)

complete $\Delta T = T_H - T_L$

Steps to solve
$\Delta u_{\text{total}} = \Delta u_1 + \Delta u_2 + \Delta u_3 + \Delta u_4$
$\Delta w_{\text{total}} = \Delta w_1 + \Delta w_2 + \Delta w_3 + \Delta w_4$ which formulas use in each step is above

stages of a reaction

ie argon changes temp and volume, we could calculate entropy of each step separately when volume changes $\Delta S = nR\ln\left(\frac{V_f}{V_i}\right)$ when temp changes

$$\Delta S = nC_{v,n}\ln\left(\frac{T_f}{T_i}\right)$$

temp exchange of two substances we calculate T equilibrium
and then we know $\Delta S = \Delta S_1 + \Delta S_2$ we sum both changues

phase changes in a reaction

for example vaporization of water this is a rev procces and assuming constant pressure we know $q = \Delta H_{\text{vaporization}}$ thus

$$\Delta S = \frac{q}{T} = \frac{\Delta H}{T}$$